

**WRITE YOUR ANSWERS ON THIS EXAM PAPER**

**PLACE THIS PAPER AND ANY ANSWER BOOKLETS IN  
THE EXAM ENVELOPE PROVIDED**

**Name of Faculty: Natural, Mathematical & Engineering Sciences**

**Name of Department: Mathematics**

**Examination Period 2 (Summer 2026)**

|                                                                                                                                                                                             |                                                                                                                                                                                                                                                      |                                                          |     |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------|-----|
| <b>Candidate Number:</b>                                                                                                                                                                    |                                                                                                                                                                                                                                                      | <b>Seat Number (including colour zone if applicable)</b> |     |
| <b>Module Code:</b>                                                                                                                                                                         | 7CCMFM04                                                                                                                                                                                                                                             |                                                          |     |
| <b>Module Title:</b>                                                                                                                                                                        | STOCHASTIC ANALYSIS:<br>OLD 2024-25 SYLLABUS                                                                                                                                                                                                         |                                                          |     |
| <b>Time Allowed:</b>                                                                                                                                                                        | TWO hours                                                                                                                                                                                                                                            | <b>Total Marks:</b>                                      | 100 |
| <b>Total Pages:</b>                                                                                                                                                                         | ?? pages                                                                                                                                                                                                                                             | <b>Total Questions:</b>                                  | 4   |
| <b>Casio Fx83 or Fx85 Calculators Permitted</b>                                                                                                                                             | No                                                                                                                                                                                                                                                   |                                                          |     |
| <b>Supplementary Booklet Required</b>                                                                                                                                                       | This exam does not require a supplementary booklet.                                                                                                                                                                                                  |                                                          |     |
| <b>Any Other Permitted Materials</b>                                                                                                                                                        | Books, notes or other written material may NOT be brought into this examination.                                                                                                                                                                     |                                                          |     |
| <b>Instructions for candidates</b>                                                                                                                                                          | This paper consists of 4 questions; you must answer all the questions in the designated answer space.<br>All questions carry equal marks. FULL MARKS will be awarded for complete answers to all FOUR questions.<br>Write in PEN on THIS exam paper. |                                                          |     |
| This paper is part of an examination of the College counting towards the award of a degree. Examinations are governed by the College Regulations under the authority of the Academic Board. |                                                                                                                                                                                                                                                      |                                                          |     |
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If you require extra space to write your answers, please use the space at the end of the booklet on pages 18-21 (clearly write the question number on the extra page and the phrase 'see Extra space Page No. XXX' in the main section).

**Q 1.** Let  $(W_t)_{t \geq 0}$  denote a standard Brownian motion on some probability space  $(\Omega, \mathcal{F}, \mathbb{P})$  with filtration  $(\mathcal{F}_t)_{t \geq 0}$

- i) For an  $(\mathcal{F}_t^W)_{t \geq 0}$ -stopping time  $\tau$ , give the definition of  $\mathcal{F}_\tau^W$ . (proof not required). **[30%]**

**Solution.** The set events  $A$  such that  $A \cap \{\tau \leq t\} \in \mathcal{F}_t$

- ii) Let

$$X_t = \int_0^t \kappa(u, t) dW_u$$

where  $\kappa \in L^2$ . Compute  $\mathbb{E}(X_t)$  and write out an integral formula for  $\mathbb{E}(X_t^2)$  (you do not need to evaluate this integral). **[40%]**

**Solution.**

$$\begin{aligned} \mathbb{E}(X_t) &= 0 \\ \mathbb{E}(X_t^2) &= \int_0^t \kappa(t-u)^2 du. \end{aligned}$$

- iii)

**Q 2.** Let  $(W_t)_{t \geq 0}$  denote a standard Brownian motion on some probability space  $(\Omega, \mathcal{F}, \mathbb{P})$  with filtration  $(\mathcal{F}_t)_{t \geq 0}$ .

- i) What is the conditional distribution of  $W_t | W_{s_1}, \dots, W_{s_n}$  where  $s_1 < \dots < s_n < t$ . **[30%]**

**Solution.**  $W_t | W_{s_1}, \dots, W_{s_n} \sim N(W_{s_n}, t - s_n)$

- ii) Let  $B_t^{(i)} = (1-t)W_{\frac{t}{1-t}}^{(i)}$  for  $t \in [0, 1)$ , where  $(W^{(i)})_{i=1..3}$  are three independent Brownian motions, and set  $R_t = \sqrt{\sum_{i=1}^3 (B_t^{(i)})^2}$ . Compute  $\mathbb{E}(R_t^2)$ . What can we say about the process  $R_t$  as  $t \rightarrow 0$ ? **[40%]**

**Solution.**

$$\mathbb{E}(R_t^2) = \mathbb{E}\left(\sum_{i=1}^3 (B_t^{(i)})^2\right) = 3(1-t)^2 \frac{t}{1-t} = 3t(1-t).$$

Hence  $R_t$  is a non-negative process on  $[0, 1)$  with  $R_0 = 0$  and  $\lim_{t \rightarrow 1} \mathbb{E}(R_t^2) = 0$ , so in particular  $R_t \rightarrow 0$  as  $t \rightarrow 1$ .

- iii) Let  $X_t = \mu t + \sigma W_t$  and  $\bar{X}_t = \max_{0 \leq s \leq t} X_s$ . Using that

$$\mathbb{P}(\bar{X}_t > b | X_t = x) = h(b) = e^{-\frac{2b(b-x)}{\sigma^2 t}}$$

for  $b \geq \max(x, 0)$ , explain how you would simulate  $\bar{X}_t$  given that  $X_t = x$ . **[30%]**

**Solution.** Set  $\bar{X}_t = h^{-1}(U)$  where  $U \sim U[0, 1]$

**Q 3.** Throughout this question, we let  $W = (W_t)_{t \geq 0}$  be a real-valued Brownian motion on a probability space  $(\Omega, \mathcal{F}, \mathbb{P})$ , and let  $M_t = \max_{0 \leq s \leq t} W_s$  denote the running maximum process of  $W$  and  $\tau_b = \inf\{t \geq 0 : W_t = b\}$ .

- i) State the reflection principle for Brownian motion and use it to compute  $\mathbb{P}(M_t \geq b, W_t \in dx)$ . **[40%]**

**Solution.**

$$\mathbb{P}(M_t \geq b, W_t \leq x) = \mathbb{P}(W_t \geq 2b - x) = \Phi^c\left(\frac{2b - x}{\sqrt{t}}\right)$$

for  $b \geq x$ . Then

$$\mathbb{P}(M_t \geq b, W_t \in dx) = n\left(\frac{2b - x}{\sqrt{t}}\right)dx$$

where  $n(\cdot)$  is the standard Normal density.

- ii) If  $M_t = a$  for  $t \in [t_1, t_2]$  and  $M_t \neq a$  otherwise, what can we say about the  $(\tau_b)_{b \geq 0}$  process at  $b = a$ ? (hint: draw a picture). **[30%]**

**Solution.**  $(\tau_b)_{b \geq 0}$  has a jump at  $b = a$  of size  $t_2 - t_1$ .

- iii) Write down an expression for the density of  $\tau_b$ .

**Solution.** See Reflection Principle chapter

**[30%]**

**Q 4.** Let  $(W_t)_{t \geq 0}$  denote a standard Brownian motion on some probability space  $(\Omega, \mathcal{F}, \mathbb{P})$  with filtration  $(\mathcal{F}_t)_{t \geq 0}$ .

i) Consider the SDE

$$dX_t = X_t^2 dW_t$$

with  $X_0 > 0$ . Derive the SDE satisfied by  $Y_t = \frac{1}{X_t}$ , in terms of  $Y_t$ . **[30%]**

**Solution.** From Ito's lemma,

$$dY_t = -\frac{1}{Y_t^2} dX_t + \frac{1}{Y_t^3} d\langle X \rangle_t = -\frac{1}{Y_t^2} X_t^2 dW_t + \frac{1}{Y_t^3} X_t^4 dt = -Y_t dt - dW_t$$

ii) Let  $S$  satisfy the SDE

$$dS_t = \sigma S_t^\beta dW_t$$

with  $S_0 > 0$  for  $\beta > 0$  (with absorption at zero), and let  $\mathcal{F}^S$  denote the natural filtration of  $S$ . Using that a general one-dimensional diffusion of the form  $dX_t = \sigma(X_t)dW_t$  is an  $\mathcal{F}^X$ -martingale if and only if  $\int_r^\infty xm(x)dx = \infty$  for all  $r > 0$  (where  $m(x) = 1/\sigma(x)^2$ ), compute the range of  $\beta$  values for which  $S$  is an  $\mathcal{F}^S$ -martingale. **[40%]**

**Solution.** We find that  $S$  is a martingale if and only if  $\beta \leq 1$ .